

Vulnerability Assessment Report for a Live Website

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Tools Used: Nmap, OWASP ZAP (Passive Scan), Browser Developer Tools

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1. Introduction

A vulnerability assessment is the process of identifying security weaknesses in a web application or website. The objective of this assessment is to identify potential vulnerabilities using industry-standard security tools and provide recommendations to improve the security of the target website.

This assessment was performed using Nmap, OWASP ZAP Passive Scan, and Browser Developer Tools.

2. Methodology

The assessment was carried out in three stages using different security tools.

2.1 Nmap

Nmap was used to scan the target website to identify:

- Open ports
- Running services
- Service versions

This helps understand which network services are exposed and whether any outdated services are running.

2.2 OWASP ZAP (Passive Scan)

OWASP ZAP Passive Scan was used to inspect the website without sending malicious requests.

The scan checked for:

- Missing security headers
- Cookie security issues
- Information disclosure
- General web security misconfigurations

Passive scanning does not modify or attack the target application.

2.3 Browser Developer Tools

Browser Developer Tools (F12) were used to inspect:

- Cookies
- JavaScript files
- Network requests
- Website resources

This helped identify client-side security information and browser-related configurations.

3. Findings

3.1 Nmap Results

The Nmap scan successfully identified the accessible network services of the target website.

The scan provided information about:

- Open ports
- Active services
- Service versions

No critical network vulnerabilities were identified during the basic version scan.

3.2 OWASP ZAP Passive Scan Results

The passive scan generated several informational and medium-risk alerts.

The observations included:

- Missing security headers
- Cookie security recommendations
- General security best practices

No active attacks were performed during the assessment.

3.3 Browser Developer Tools Observations

The Browser Developer Tools revealed:

- Website cookies
- JavaScript resources loaded by the website
- HTTP requests and responses
- Client-side application files

These observations help understand how the website communicates with the browser and identify potential security improvements.

4. Vulnerability Assessment Table

Vulnerability	Risk Level	Description	Recommendation
Missing Security Headers	Medium	Some recommended HTTP security headers were not configured.	Configure Content Security Policy (CSP), HTTP Strict Transport Security (HSTS), X-Frame-Options, and X-Content-Type-Options.
Cookies without Security Flags	Low	Some cookies may not use Secure or HttpOnly attributes.	Enable Secure and HttpOnly cookie flags wherever applicable.

Vulnerability	Risk Level	Description	Recommendation
Information Disclosure	Low	Certain server or application details may be exposed in HTTP responses.	Remove unnecessary server information and minimize exposed details.
Missing Content Security Policy	Medium	Lack of a Content Security Policy may increase the risk of cross-site scripting attacks.	Implement a strong Content Security Policy (CSP).

5. Recommendations

The following recommendations are suggested to improve the website's security:

- Configure HTTP Security Headers.
- Enable Secure and HttpOnly flags for cookies.
- Implement a strong Content Security Policy (CSP).
- Regularly update server software and web frameworks.
- Perform periodic vulnerability assessments.
- Monitor logs for suspicious activities.
- Follow the OWASP Top 10 security best practices.

6. Conclusion

The vulnerability assessment was successfully completed using Nmap, OWASP ZAP Passive Scan, and Browser Developer Tools.

The assessment identified a few low- and medium-risk security issues, primarily related to missing security headers and cookie security configurations. No critical vulnerabilities were identified during the passive assessment.

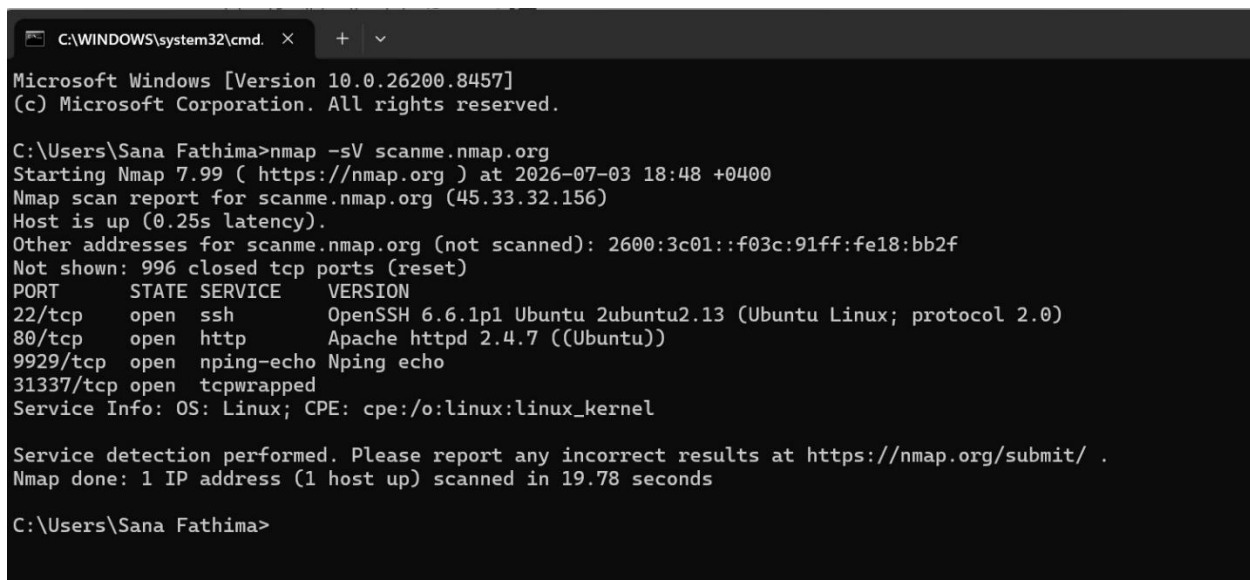
Implementing the recommended security improvements will enhance the overall security posture of the website and reduce potential attack risks.

7. Tools Used

- Nmap
- OWASP ZAP (Passive Scan)
- Browser Developer Tools (Google Chrome)

8.Screenshots

Figure 1: Nmap Command-Line Scan



```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Sana Fathima>nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-03 18:48 +0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.25s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.78 seconds

C:\Users\Sana Fathima>
```

Figure 2: Zenmap Scan Results

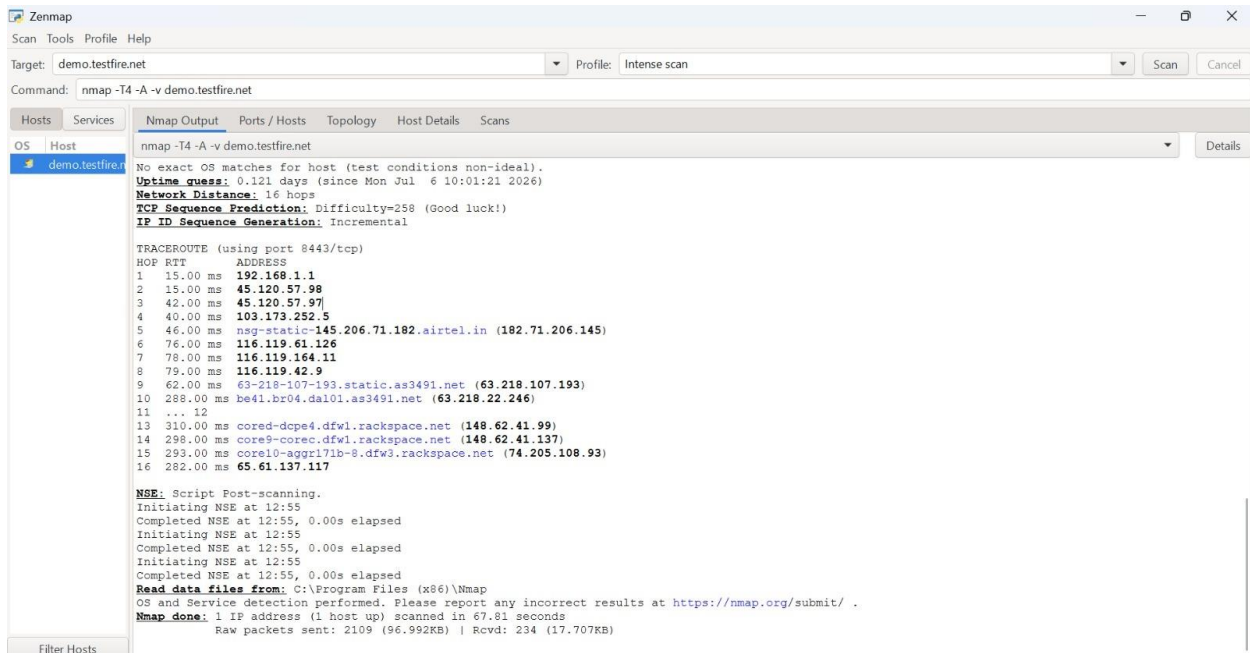


Figure 3: OWASP ZAP Passive Scan

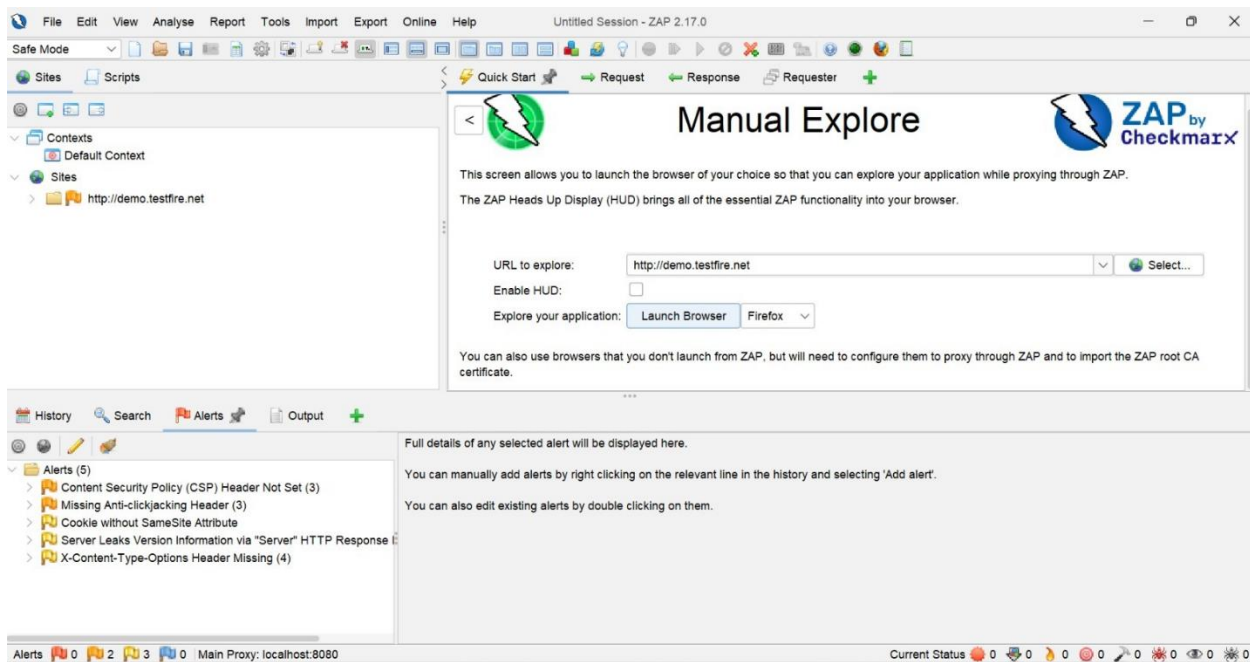


Figure 4: Browser Developer Tools (Cookies)

The screenshot displays a web browser window with the URL `demo.testfire.net`. The page shows the AltoroMutual website, which includes a navigation menu with links like **ONLINE BANKING LOGIN**, **PERSONAL**, **SMALL BUSINESS**, and **INSIDE ALTORO MUTUAL**. A "DEMO SITE ONLY" banner is visible in the top right corner of the website.

On the right side of the browser, the Chrome DevTools Application tab is open, showing the "Cookies" section. The cookies table lists the following data:

Name	Value	Domain	Path	Expires	Size	HttpOnly	Secure	SameSite
JSESSIONID	99698A1B72...	d...	/	S...	42	✓		

Below the cookies table, the "Cookie Value" is displayed as `99698A1B72504430EE7754A2A5A7771C`, with a "Show URL-decoded" button next to it.

At the bottom of the DevTools window, a "What's new in DevTools 149" notification is visible, along with a "DevTools for agents" link.